

boxes, labels, menus, buttons and common elements.

- **Physics engine:** It supports 2D physics engines like BOX2D and Chipmunk.
- **Audio:** It supports sound effects and background music.
- **Network support:** HTTP with SSL, WebSocket API, XMLHttpRequest API, etc.

Latest version: 3.15.1

Official website: <http://www.cocos2d-x.org>

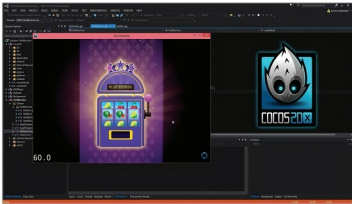


Figure 3: Cocos2d-x user interface

Delta Engine

Delta Engine is an open source 2D and 3D app and game development engine maintained by Delta Engine company. The applications and games can be developed in an easy manner through *Visual Studio.net* or the Delta engine editor. Delta Engine supports various languages and frameworks like C# OpenGL, C# OpenTK, C# GLFW, C# XNA, C# SharpDX, C# SlimDX, C# MonoGame, LinuxOpenGL, MacOpenGL and WebGL.

It supports various platforms like Windows OS, OS X, Linux, Android, Android TV and Linux.

Features

- It supports 3D features like 3D model importing, a particle effect editor, etc.
- Content like images, sounds, music and 3D models is saved directly using the Delta engine.
- Supports physical simulation; most code is interchangeable for both 2D and 3D simulation.
- Supports integration of external libraries and frameworks like the 2D Sprite animation library, Spine.
- App Builder tool integrated in the editor supports building, deployment and launching of apps on a mobile device.

Latest version: 0.9.11

Official website: <https://deltaengine.net>

Starling

Starling is an open source 2D game development framework that supports both mobile and desktop platforms. It is a pure

ActionScript 3 library that is very similar to the traditional Flash architecture. It recreates the Flash display list architecture on top of the accelerated graphics hardware.

It is a very compact framework but comprises various packages and classes. The following are the sets of tools that are integrated with Starling for application development:

- **Display programming:** Every object is a display object.
- Images and textures
- Dynamic text
- Event handling
- Animation
- Asset management
- Special effects
- Utilities

Features

- It is based on Stage3D and supports multiple platforms like Android, iOS, Web browsers, OS X, etc.
- It has low configuration requirements in terms of CPU, memory and GPU.
- It has lower battery consumption.
- Has effective object organisation via hierarchical trees, i.e., parent-child relationship.
- Highly powerful and efficient event system using ActionScript.
- Supports texture atlases, filters, stencil masks, blend modes, Tweens, multi-touch, bitmap fonts and 3D effects.

Latest version: 2.2

Official website: <https://gamua.com/starling>

Panda 3D

Panda 3D is an open source framework for rendering and developing 3D games using C++ and Python programs. The entire gaming engine is written in C++ and makes use of automatic wrapper generators to expose the complete functionality of the engine in the Python interface. It supports OpenGL and DirectX.

Panda 3D includes various tools like scene graph browsing, performance monitoring, animation optimisers and many more.

Features

- Hassle-free installation and supports Windows, OS X and Linux. No need for any sort of compilation.
- Full Python integration and highly optimised via C++.
- Comes with various OpenGL and DirectX features like GLSL, a powerful interface between shaders and engine, and supports render-to-texture and multiple render targets.
- Other features include shader generation, 3D pipeline, support for OpenAL Audio Engine, FMOD Audio Engine and Miles Audio Engine.
- Has support for the Bullet physics engine, ODE physical engine and PhysX physics engine.